

Arraylists

Arrays → any disadvantage?

- fixed size (we can't increase/decrease size of array)
 - Can't remove element randomly
 - Size should be known in advance.
-
- Shopping list
 - News tabs in browser
 - ⋮

Arraylist

all the advantages of array plus some additional features.

- Dynamic Size
- No need to know the size in advance

Syntax

`ArrayList <Type> arr = new ArrayList <Type>();`

↓
has to be class,
can't be primitive

zero-size
arraylist

primitive : int, float, double, boolean ...

Java has wrapper classes for all
primitive types.

Integer, Long, Double, String, Boolean ...

Basic Operations

Insert

`add()` ⇒ add element at the end of list

Get

`get(index)` ⇒ fetch element at given index

Size

`size()` ⇒ size of arraylist

Remove

`remove(index)` $\Rightarrow$ remove element at given index.

Set

`set(index, newVal)` $\Rightarrow$ update the index with new value.

Question 1

Given an ArrayList as input, return an ArrayList of multiples of 5 or 7.

i/p : [1, 5, 3, 0, 7]

o/p : [5, 0, 7]

```
ArrayList<Integer> multiples(ArrayList<Integer> arr){
```

```
    ArrayList<Integer> ans = new ArrayList<Integer>();
```

```
    for(int i=0; i<arr.size(); i++) {
```

```
        int val = arr.get(i);
```

```
if (val % 5 == 0 || val % 7 == 0)
    ans.add(val);
```

```
}
```

```
return ans;
```

```
}
```

Question 2

Given 2 integers A & B as input, return an ArrayList containing first B multiples of A.

i/p : A = 2, B = 4

o/p : [2, 4, 6, 8]

Solution : $A \times 1, A \times 2, A \times 3, \dots, A \times B$

```
ArrayList<Integer> firstB (int A, int B) {
```

```
    ArrayList<Integer> ans = new ArrayList<Integer>();
```

```
    for (int i = 1; i <= B; i++) {
        ans.add (A * i);
    }
```

L
}

return am;

}

OR

for (int i = A; i <= A * B; i += A) {

am.add(i);

}

2D-Array → array of arrays

2D-Arraylist → arraylist of arraylists

Syntax

`ArrayList<ArrayList<Type>> mat = new ArrayList<ArrayList<Type>>();`

-Add : adds arraylist inside 2D-Arraylist.

we can have different size ArrayList in a single 2D Arraylist.

Question

Given integer N as input, return a numeric staircase.

i/p : 3

o/p : $\begin{bmatrix} [1], \\ [1, 2], \\ [1, 2, 3] \end{bmatrix}$

4
 $\begin{bmatrix} [1], \\ [1, 2], \\ [1, 2, 3], \\ [1, 2, 3, 4] \end{bmatrix}$

```
ArrayList<ArrayList<Integer>> staircase (int N) {
```

```
    ArrayList<ArrayList<Integer>> ans = new ArrayList<  
        ArrayList<Integer>>();
```

```
    for (int row = 1 ; row <= N ; row++) {
```

```
        ArrayList<Integer> rw = new ArrayList<Integer>();
```

```
        for (int col = 1 ; col <= row ; ++col) {
```

```
            rw.add(col);
```

```
        }
```

```
        ans.add(rw);
```

```
}
```

return am;

}

Use Java 8 Oracle JDK for ArrayList

[https://www.scaler.com/topics/java/online-java-compiler/?
snippet_slug=4cd9277c7ed48058fe95](https://www.scaler.com/topics/java/online-java-compiler/?snippet_slug=4cd9277c7ed48058fe95)

[https://www.scaler.com/topics/java/online-java-compiler/?
snippet_slug=fdc7753c0df90118bac0](https://www.scaler.com/topics/java/online-java-compiler/?snippet_slug=fdc7753c0df90118bac0)